

# Logarithmic Uncertainty Relations for Odd or Even Signals Associate with Wigner–Ville Distribution

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**Abstract** Heisenberg’s uncertainty relation is a basic principle in the applied mathematics and signal processing community. The logarithmic uncertainty relation, which is a more general form of Heisenberg’s uncertainty relation, describes the relationship between a function and its Fourier transform. In this paper, we consider several logarithmic uncertainty relations for a odd or even signal  $f(t)$  related to the Wigner–Ville distribution and the linear canonical transform. First, the logarithmic uncertainty relations associated with the Wigner–Ville distribution of a signal  $f(t)$  based on the Fourier transform are obtained. We then generalize the logarithmic uncertainty relation to the linear canonical transform domain and derive a number of theorems relating to the Wigner–Ville distribution and the ambiguity function; finally, the logarithmic uncertainty relations are obtained for the Wigner–Ville distribution associated with

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the linear canonical transform. We present an example in which the theorems derived in this paper can be used to provide an estimation for a practical signal.

**Keywords** Logarithmic uncertainty relation · Wigner–Ville distribution · Linear canonical transform · Fourier transform

## 1 Introduction

Heisenberg’s uncertainty principle is of considerable importance in the field of signal processing and plays a very important role in quantum mechanics, applied mathematics, physics, and optics [11, 12, 19]. The principle implies that a nonzero function and its Fourier transform cannot both be sharply localized. The logarithmic uncertainty relation can be considered as a more general form of Heisenberg’s uncertainty relation. Thus, it is meaningful to study the logarithmic uncertainty relation.

The recently developed linear canonical transform (LCT) has three parameters and can be thought of as an extension of the classical Fourier transform [20, 23]. The LCT reflects an affine transformation in the time–frequency plane. Its rapid development [14, 22, 23] has seen the publication of many interesting results associated with the LCT [1, 3, 5, 8, 14, 16, 18, 20, 22, 23, 25, 26, 28]. Uncertainty principles associated with the LCT have also been investigated [8, 18, 25–28].

Time–frequency analysis is an important mathematical tool for analyzing nonstationary random signals. The Wigner–Ville distribution (WVD) is the basis of time–frequency analysis distributions, because other time–frequency distributions are essentially versions of the WVD with added window functions. In recent years, Pei and Ding investigated the definition and properties of WVDs associated with the LCT [16]. Bai later proposed another definition of the WVD associated with the LCT (WDL) [1]. The WVD has an important role in signal analysis and processing, especially for nonstationary signals [1, 3, 5, 7, 16]. Thus, it is essential to study the uncertainty principles associated with the WVD, and hence Heisenberg’s inequality for the WVD has been investigated [2, 7, 10, 13].

The logarithmic uncertainty relation in the Fourier domain [10] plays an important role in applied mathematics. This makes the logarithmic uncertainty relation for the WVD worthy of investigation. However, to the best of our knowledge, no such studies have been published in the literature. Therefore, it is interesting and worthwhile to study this type of uncertainty relation. In this paper, we will investigate the logarithmic uncertainty relation for the WVD on the basis of the logarithmic uncertainty relation in the Fourier domain and investigate related results in the LCT domain.

The remainder of this paper is organized as follows. In Section 2, we introduce some general definitions and present certain conclusions that are useful for later theorems. In Section 3, we first obtain the logarithmic uncertainty relation for the WVD of a signal  $f(t)$  and then derive the logarithmic uncertainty relation associated with the LCT. Finally, we obtain the logarithmic uncertainty relation for the WVD associated with the LCT. Our conclusions are summarized in Section 4.

## 2 Preliminaries

### 2.1 The Uncertainty Principles

Heisenberg's uncertainty principle states that the complementary variables of microscopic particles, such as their position and momentum or time and energy, cannot simultaneously have prescribed values. That is, if we try to limit the behavior of one variable, we lose control of the other. The logarithmic uncertainty relation can be seen as a variant of Heisenberg's uncertainty principle, and the logarithmic uncertainty relation associated with the Fourier transform has been derived as follows [2, 6, 21].

**Lemma 1** (Logarithmic uncertainty relation of a Fourier transform [2]) *For  $f \in T(\mathbb{R}^n)$ ,*

$$\int_{\mathbb{R}^n} |f(x)|^2 \ln |x| \, dx + \int_{\mathbb{R}^n} |\widehat{f}(\xi)|^2 \ln |\xi| \, d\xi \geq D \int_{\mathbb{R}^n} |f(x)|^2 \, dx,$$

where  $D = \psi(\frac{n}{4}) - \ln \pi$ ,  $\psi(t) = \frac{d}{dt} \ln [\Gamma(t)]$ , and  $\Gamma(t)$  is the Gamma function. Here,  $T(\mathbb{R}^n)$  denotes the Schwartz class and  $\widehat{f}$  is the Fourier transform. When

$$\int_{\mathbb{R}^n} |\widehat{f}(\xi)|^2 |\xi|^{-\alpha} \, d\xi = D_\alpha \int_{\mathbb{R}^n} |f(x)|^2 |x|^\alpha \, dx,$$

the inequality becomes an equality, where  $D_\alpha = \frac{\pi^\alpha \Gamma^2(\frac{n-\alpha}{4})}{\Gamma^2(\frac{n+\alpha}{4})}$ ,  $0 \leq \alpha < n$ .

When  $n = 2$ ,  $f(x) \in L^2(\mathbb{R})$ , and  $\|f\|_2 = 1$ , we can obtain the logarithmic uncertainty relation in the two-dimensional Fourier transform domain as follows:

$$\begin{aligned} & \int_{\mathbb{R}^2} |f(x, y)|^2 \ln \sqrt{x^2 + y^2} \, dx \, dy + \int_{\mathbb{R}^2} |\widehat{f}(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv \\ & \geq \psi\left(\frac{1}{2}\right) - \ln \pi. \end{aligned} \quad (1)$$

This inequality is realized as an a priori limit, because the formula on the left-hand side of the above expression may be indeterminate on  $L^2(\mathbb{R})$ , and when

$$\int_{\mathbb{R}^2} |\widehat{f}(u, v)|^2 (u^2 + v^2)^{-\frac{\alpha}{2}} \, du \, dv = D_\alpha \int_{\mathbb{R}^2} |f(x, y)|^2 (x^2 + y^2)^{\frac{\alpha}{2}} \, dx \, dy,$$

the inequality becomes an equality, where  $D_\alpha = \frac{\pi^\alpha \Gamma^2(\frac{2-\alpha}{4})}{\Gamma^2(\frac{2+\alpha}{4})}$ ,  $0 \leq \alpha < 2$ . (See [2] for more details about this lemma.)

### 2.2 Time–Frequency Analysis

In signal processing, time–frequency analysis refers to techniques that study a signal in the time and frequency domains simultaneously, using various time–frequency

representations. The WVD and the ambiguity function (AF) are important tools for time–frequency analyses associated with the classical Fourier domain. The WVD of signals  $f$  and  $g$  is defined as [6]:

$$W(f, g)(t, u) = \int_{\mathbb{R}} e^{-2\pi i u \tau} f\left(t + \frac{\tau}{2}\right) g^*\left(t - \frac{\tau}{2}\right) d\tau, \quad (2)$$

where  $g^*(t)$  is the conjugate function.

The AF of the signals  $f$  and  $g$  is defined as [6]:

$$A(f, g)(\tau, u) = \int_{\mathbb{R}} e^{2\pi i u t} f\left(t + \frac{\tau}{2}\right) g^*\left(t - \frac{\tau}{2}\right) dt. \quad (3)$$

From their definitions, we can see that these are both bilinear transformations of the signal. In the time–frequency domain, the AF represents correlation, whereas the WVD can localize the signal energy in both the time and frequency directions. It is easy to show that the WVD is equivalent to the two-dimensional Fourier transform of the AF, i.e.,

$$W(f, g)(t, u) = \int_{\mathbb{R}^2} A(f, g)(\tau, v) e^{-2\pi i (u\tau + vt)} dv d\tau. \quad (4)$$

**Lemma 2** Suppose that  $W(f, g)$  is the WVD of  $f(t)$  and  $g(t)$ , and  $A(f, g)$  is the AF of  $f(t)$  and  $g(t)$ . Then it is shown in [7, 17]:

$$W(f, g)(\xi, x) = 2A(f, \tilde{g})(2x, -2\xi), \quad (5)$$

where  $[\tilde{g}(x) = g(-x)]$ .

In fact, if a signal  $f(t)$  is an odd or even function, then

$$W_f(\xi, x) = -2A_f(2x, -2\xi)$$

and

$$W_f(\xi, x) = 2A_f(2x, -2\xi).$$

**Lemma 3** (Moyal formula [6, 21]): For  $f(t) \in L^2(\mathbb{R})$ , the following equality holds:

$$\int_{\mathbb{R}^2} |W_f(t, u)|^2 dt du = \left( \int_{\mathbb{R}} |f(t)|^2 dt \right)^2. \quad (6)$$

This lemma can be easily proved using the definitions of each quantity. For more details, see [6].

### 2.3 The Linear Canonical Transform

The LCT is a family of integral transforms that generalize many classical transforms, such as the Fourier transform and Fractional Fourier transform. The one-dimensional LCT [23] of a signal  $f(t)$  can be written as

$$F_A(u) = [\mathcal{C}_A f(t)](u) = \begin{cases} \int_{\mathbb{R}} f(t) K_A(u, t) dt, & b \neq 0 \\ \sqrt{d} e^{\frac{i}{2} c d u^2} f(du), & b = 0 \end{cases} \quad (7)$$

where  $K_A(u, t) = \sqrt{\frac{1}{i2\pi b}} e^{i \frac{a}{2b} t^2 - i \frac{1}{b} ut + i \frac{d}{2b} u^2}$ , and  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  is the parameter matrix of the LCT satisfying  $|A| = ad - bc = 1$ .

The two-dimensional LCT (2D LCT) [24] of a function  $f(x; y) ((x; y) \in \mathbb{R}^2)$  can be defined as follows:

$$F_{A,B}(u, v) = \int_{\mathbb{R}^2} f(x, y) K_{A,B}(u, v, x, y) dx dy, \quad (8)$$

where  $K_{A,B}(u, v, x, y) = K_A(u, x) K_B(v, y)$ , and  $A = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ ,  $B = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$  satisfy  $|A| = |B| = 1$ .

The LCT has a series of useful properties. Those that are relevant to this article are listed below.

- (1) Let  $F_A(u)$  be the LCT of  $f(t)$  with parameter  $A$ . If  $f(t)$  is an odd function, then  $F_A(u)$  is odd; if  $f(t)$  is an even function, then  $F_A(u)$  is even.
- (2) (Parseval Principle)

$$\int_{\mathbb{R}} |f(t)|^2 dt = \int_{\mathbb{R}} |F_A(u)|^2 du. \quad (9)$$

With the rapid development of the LCT in the signal processing and mathematical communities, the relationship between the LCT and time–frequency distributions have been extensively investigated in the literature. Pei and Ding investigated the AF and WVD associated with the LCT [16]. They defined the AF for the LCT as

$$A_F(\tau, \omega) = \int_{\mathbb{R}} F_A\left(u + \frac{\tau}{2}\right) F_A^*\left(u - \frac{\tau}{2}\right) e^{2\pi i \omega u} du,$$

and the WVD associated with the LCT as

$$W_F(u, \omega) = \int_{\mathbb{R}} F_A\left(u + \frac{\tau}{2}\right) F_A^*\left(u - \frac{\tau}{2}\right) e^{-2\pi i \omega \tau} d\tau,$$

where  $F_A(u)$  is the LCT of  $f(t)$  and  $F_A^*(u)$  is the conjugate function.

The relationship between this kind of WVD and AF can be derived as

$$W_F(u, \omega) = \int_{\mathbb{R}^2} A_F(\tau, v) e^{-2\pi i(\omega\tau + uv)} \, d v \, d \tau. \quad (10)$$

Different from the definition in [16], Bai proposed a new WVD associated with the LCT (the so-called WDL)[1] as follows:

$$W_A^f(t, u) = \sqrt{\frac{1}{i2\pi b}} \int_{\mathbb{R}} f\left(t + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) e^{i\frac{a}{2b}\tau^2 - i\frac{1}{b}u\tau + i\frac{d}{b}u^2} \, d \tau.$$

Similarly, Che defined the AF associated with the LCT (AFL)[4] as

$$AFL_A^f(\tau, u) = \sqrt{\frac{1}{i2\pi b}} \int_{\mathbb{R}} f\left(t + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) e^{i\frac{a}{2b}t^2 - i\frac{1}{b}ut + i\frac{d}{b}u^2} \, d t.$$

From these definitions of WDL and AFL, we have the following equality [23] :

$$W_A^f(t, u) = \int_{\mathbb{R}^2} AFL_A^f(\tau, v) K_{A,A^{-1}}(\tau, v, u, t) \, d v \, d \tau, \quad (11)$$

where  $K_{A,A^{-1}}(\tau, v, u, t) = K_A(\tau, u) K_{A^{-1}}(v, t)$ . More results and properties for the WDL and AFL can be found in [23].

## 3 Main Results

### 3.1 Logarithmic Uncertainty Relation for the WVD

Lemma 1 showed that the classical logarithmic uncertainty relation is associated with the Fourier transform and can be considered as a variant of the classical Heisenberg uncertainty relation. Based on these existing results, we investigate the logarithmic uncertainty relation for the WVD of a signal  $f(t)$ . The main result is given in Theorem 1, and states that, if the signal is an odd or even function and its energy is 1, then the logarithmic integral of WVD is not less than some constant.

**Theorem 1** *Let  $f(t) \in L^2(\mathbb{R})$ ,  $\|f\|_2 = 1$ . If  $f(t) = f(-t)$  or  $f(t) = -f(-t)$  is satisfied, then the following inequality holds:*

$$\int_{\mathbb{R}^2} |W_f(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} \, d \tau \, d v \geq \frac{1}{2} \left[ \psi\left(\frac{1}{2}\right) - \ln 2\pi \right], \quad (12)$$

where  $\psi(t) = \frac{d}{dt} \ln [\Gamma(t)]$ , and  $\Gamma(t)$  is the Gamma function. When

$$\int_{\mathbb{R}^2} |W_f(\tau, v)|^2 (\tau^2 + v^2)^{-\frac{\alpha}{2}} \, d \tau \, d v = D_\alpha \int_{\mathbb{R}^2} |A_f(t, u)|^2 (t^2 + u^2)^{\frac{\alpha}{2}} \, d t \, d u,$$

the inequality becomes an equality, where  $D_\alpha = \frac{\pi^\alpha \Gamma^2(\frac{2-\alpha}{4})}{\Gamma^2(\frac{2+\alpha}{4})}$ ,  $0 \leq \alpha < 2$ .

*Proof* First, the following inequality can be obtained from Lemma 1:

$$\begin{aligned} & \int_{\mathbb{R}^2} |A_f(t, u)|^2 \ln \sqrt{\tau^2 + u^2} d\tau du + \int_{\mathbb{R}^2} |\widehat{A}_f(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} d\tau dv \\ & \geq \psi\left(\frac{1}{2}\right) - \ln \pi. \end{aligned} \quad (13)$$

Because the WVD is the Fourier transform of the AF, we have

$$\begin{aligned} & \int_{\mathbb{R}^2} |A_f(t, u)|^2 \ln \sqrt{t^2 + u^2} dt du + \int_{\mathbb{R}^2} |W_f(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} d\tau dv \\ & \geq \psi\left(\frac{1}{2}\right) - \ln \pi. \end{aligned}$$

When  $f(x) = f(-x)$  or  $f(x) = -f(-x)$ , we obtain

$$\begin{aligned} & \int_{\mathbb{R}^2} |W_f(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} d\tau dv + \int_{\mathbb{R}^2} \left| \frac{1}{2} W_f\left(-\frac{u}{2}, \frac{t}{2}\right) \right|^2 \ln \sqrt{t^2 + u^2} dt du \\ & \geq \psi\left(\frac{1}{2}\right) - \ln \pi. \end{aligned}$$

Let  $-\frac{u}{2} = x$ ,  $\frac{t}{2} = y$ , and substitute these in the above equation. We then obtain

$$\begin{aligned} & \int_{\mathbb{R}^2} |W_f(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} d\tau dv + \int_{\mathbb{R}^2} |W_f(x, y)|^2 \left( \ln 2 + \ln \sqrt{x^2 + y^2} \right) dx dy \\ & \geq \psi\left(\frac{1}{2}\right) - \ln \pi. \end{aligned}$$

From Lemma 3 and  $\|f\|_2 = 1$ , we then obtain the final result.

$$\int_{\mathbb{R}^2} |W_f(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} d\tau dv \geq \frac{1}{2} \left[ \psi\left(\frac{1}{2}\right) - \ln 2\pi \right]. \quad (14)$$

We can derive the conditions for the above inequality (Eq. 14) becomes an equality based on the results of Lemma 1 and Eq. 13.  $\square$

Because the WVD can be considered as the two-dimensional Fourier transform, we can derive a similar logarithmic uncertainty relation for the AF of a signal  $f(t)$  in the same manner. This is stated without proof in the following corollary. This result implies that, if the signal is an odd or even function and its energy is 1, then the logarithmic integral of the AFL is not less than some constant.

**Corollary 1** Let  $f(t) \in L^2(\mathbb{R})$  and  $\|f\|_2 = 1$ . When  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , the following inequality holds:

$$\int_{\mathbb{R}^2} |A_f(\tau, u)|^2 \ln \sqrt{\tau^2 + u^2} \, d\tau \, du \geq \frac{1}{2} \left[ \psi\left(\frac{1}{2}\right) - \ln \pi + \ln 2 \right], \quad (15)$$

where  $\psi(t) = \frac{d}{dt} \ln [\Gamma(t)]$ , and  $\Gamma(t)$  is the Gamma function. When

$$\int_{\mathbb{R}^2} |W_f(\tau, v)|^2 (\tau^2 + v^2)^{-\frac{\alpha}{2}} \, d\tau \, dv = D_\alpha \int_{\mathbb{R}^2} |A_f(t, u)|^2 (t^2 + u^2)^{\frac{\alpha}{2}} \, dt \, du,$$

the inequality becomes an equality.

### 3.2 Logarithmic Uncertainty Relation Associated with the LCT

Because the LCT is a generalization of the classical Fourier transform, we now investigate the related logarithmic uncertainty relations associated with the LCT.

First, the logarithmic uncertainty relation related to the 2D LCT can be derived from the following theorem. This result states that the logarithmic integral of  $f(x, y)$  and its 2D LCT is not less than some number with parameter  $b$ .

**Theorem 2** Let  $\|f(x, y)\|_2 = 1$  and  $F_{A,B}$  be the 2D LCT of  $f(x, y)$ , where  $A = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ ,  $B = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$  are parameter matrices satisfying  $|A| = |B| = 1$ . If  $|b_1| = |b_2| = b$ , then

$$\begin{aligned} & \int_{\mathbb{R}^2} |f(x, y)|^2 \ln \sqrt{x^2 + y^2} \, dx \, dy + \int_{\mathbb{R}^2} |F_{A,B}(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv \\ & \geq 4\pi^2 b^2 \left( \psi\left(\frac{1}{2}\right) - \ln \pi \right) + \ln 2\pi |b|. \end{aligned} \quad (16)$$

When

$$\begin{aligned} & (2\pi b)^a D_\alpha \int_{\mathbb{R}^2} |f(x, y)|^2 (x^2 + y^2)^{\frac{\alpha}{2}} \, dx \, dy \\ & = \int_{\mathbb{R}^2} |F_{A,B}(u, v)|^2 (u^2 + v^2)^{-\frac{\alpha}{2}} \, du \, dv \end{aligned}$$

the inequality becomes an equality, where  $D_\alpha = \frac{\pi^\alpha \Gamma^2(\frac{2-\alpha}{4})}{\Gamma^2(\frac{2+\alpha}{4})}$ ,  $0 \leq \alpha < 2$ .

**Proof** Based on the definition of the 2D LCT, we can derive

$$\begin{aligned} |F_{A,B}(u, v)| &= \frac{1}{2\pi} \left| \frac{1}{\sqrt{-b_1 b_2}} \int_{\mathbb{R}^2} f(x, y) e^{i \frac{a_1}{2b_1} x^2 - i \frac{1}{b_1} ux + i \frac{a_2}{2b_2} y^2 - i \frac{1}{b_2} vy} \, dx \, dy \right| \\ &= \frac{1}{2\pi} \left| \frac{1}{\sqrt{-b_1 b_2}} \int_{\mathbb{R}^2} \left( f(x, y) e^{i \frac{a_1}{2b_1} x^2 + i \frac{a_2}{2b_2} y^2} \right) e^{-i \frac{1}{b_1} ux - i \frac{1}{b_2} vy} \, dx \, dy \right|. \end{aligned}$$

Let  $g(x, y) = f(x, y)e^{i\frac{a_1}{2b_1}x^2 + i\frac{a_2}{2b_2}y^2}$  and  $x = 2\pi b_1 x_1$ ,  $y = 2\pi b_2 y_1$ . Then, we obtain

$$\begin{aligned}|F_{A,B}(u, v)| &= \left| 2\pi \sqrt{b_1 b_2} \int_{\mathbb{R}^2} g(2\pi b_1 x_1, 2\pi b_2 y_1) e^{-i2\pi(ux_1 + vy_1)} dx_1 dy_1 \right| \\ &= \left| 2\pi \sqrt{b_1 b_2} \widehat{g}(2\pi b_1 x_1, 2\pi b_2 y_1) \right|.\end{aligned}$$

From the logarithmic uncertainty relation associated with the Fourier transform in Eq. 1, we have

$$\begin{aligned}\int_{\mathbb{R}^2} |g(2\pi b_1 x_1, 2\pi b_2 y_1)|^2 \ln \sqrt{x_1^2 + y_1^2} dx_1 dy_1 + \int_{\mathbb{R}^2} |\widehat{g}(u, v)|^2 \ln \sqrt{u^2 + v^2} du dv \\ \geq \psi\left(\frac{1}{2}\right) - \ln \pi.\end{aligned}\quad (17)$$

Let  $C = \frac{1}{4\pi^2 |b_1 b_2|}$ , then the left-hand side of the above equation can be rewritten as

$$\begin{aligned}&\int_{\mathbb{R}^2} |g(2\pi b_1 x_1, 2\pi b_2 y_1)|^2 \ln \sqrt{x_1^2 + y_1^2} dx_1 dy_1 \\ &\quad + \int_{\mathbb{R}^2} |\widehat{g}(u, v)|^2 \ln \sqrt{u^2 + v^2} du dv \\ &= C \left( \int_{\mathbb{R}^2} |g(x, y)|^2 \ln \sqrt{\left(\frac{x}{2\pi b_1}\right)^2 + \left(\frac{y}{2\pi b_2}\right)^2} dx dy \right. \\ &\quad \left. + \int_{\mathbb{R}^2} |F_{A,B}(u, v)|^2 \ln \sqrt{u^2 + v^2} du dv \right) \\ &= C \left( \int_{\mathbb{R}^2} |f(x, y)|^2 \ln \sqrt{\left(\frac{x}{2\pi b_1}\right)^2 + \left(\frac{y}{2\pi b_2}\right)^2} dx dy \right. \\ &\quad \left. + \int_{\mathbb{R}^2} |F_{A,B}(u, v)|^2 \ln \sqrt{u^2 + v^2} du dv \right).\end{aligned}$$

Substituting  $|b_1| = |b_2| = b$  into the above equation gives the final result:

$$\begin{aligned}\int_{\mathbb{R}^2} |f(x, y)|^2 \ln \sqrt{x^2 + y^2} dx dy + \int_{\mathbb{R}^2} |F_{A,B}(u, v)|^2 \ln \sqrt{u^2 + v^2} du dv \\ \geq 4\pi^2 b^2 \left( \psi\left(\frac{1}{2}\right) - \ln \pi \right) + \ln 2\pi |b|.\end{aligned}$$

We can derive the conditions for the above inequality becomes an equality based on the results of Lemma 1 and Eq. 17.  $\square$

Second, we derive the logarithmic uncertainty relation associated with the newly proposed WVD associated with the LCT in [16]. The result indicates that, if the signal

is an odd or even function and its energy is 1, then the logarithmic integral of the WVD associated with the LCT is not less than some constant.

**Theorem 3** Suppose  $F_A(u)$  is the LCT of  $f(t)$  with parameter matrix  $A$ , and  $W_F$  is the WVD of  $F_A(u)$  proposed in [16]. If  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , we have

$$\int_{\mathbb{R}^2} |W_F(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} \, d\tau \, dv \geq \frac{1}{2} \left[ \psi \left( \frac{1}{2} \right) - \ln 2\pi \right]. \quad (18)$$

When

$$\int_{\mathbb{R}^2} |W_F(\tau, v)|^2 (\tau^2 + v^2)^{-\frac{\alpha}{2}} \, d\tau \, dv = D_\alpha \int_{\mathbb{R}^2} |A_F(t, u)|^2 (t^2 + u^2)^{\frac{\alpha}{2}} \, dt \, du,$$

the inequality becomes an equality, where  $D_\alpha = \frac{\pi^\alpha \Gamma^2(\frac{2-\alpha}{4})}{\Gamma^2(\frac{2+\alpha}{4})}$ ,  $0 \leq \alpha < 2$ .

*Proof* First, we obtain

$$\begin{aligned} & \int_{\mathbb{R}^2} |A_F(u, \omega)|^2 \ln \sqrt{u^2 + \omega^2} \, du \, d\omega + \int_{\mathbb{R}^2} |\widehat{A}_F(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} \, d\tau \, dv \\ & \geq \psi \left( \frac{1}{2} \right) - \ln \pi. \end{aligned}$$

Then, because  $W_F$  is the Fourier transform of  $A_F$ , we have

$$\begin{aligned} & \int_{\mathbb{R}^2} |A_F(u, \omega)|^2 \ln \sqrt{u^2 + \omega^2} \, du \, d\omega + \int_{\mathbb{R}^2} |W_F(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} \, d\tau \, dv \\ & \geq \psi \left( \frac{1}{2} \right) - \ln \pi. \end{aligned}$$

When  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , we have  $F_A(u) = F_A(-u)$  or  $F_A(u) = -F_A(-u)$ . Thus, from Lemma 2, the following inequality holds:

$$\begin{aligned} & \int_{\mathbb{R}^2} \left| \frac{1}{2} W_F \left( \frac{u}{2}, \frac{\omega}{2} \right) \right|^2 \ln \sqrt{u^2 + \omega^2} \, du \, d\omega + \int_{\mathbb{R}^2} |W_F(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} \, d\tau \, dv \\ & \geq \psi \left( \frac{1}{2} \right) - \ln \pi. \end{aligned}$$

From Parseval Principle and Lemma 3, we have

$$\int_{\mathbb{R}^2} |W_F(\tau, v)|^2 \, d\tau \, dv = \int_{\mathbb{R}} |F_A(u)|^2 \, du = \int_{\mathbb{R}} |f(t)|^2 \, dt = 1.$$

Thus, we obtain

$$\int_{\mathbb{R}^2} |W_F(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} \, d\tau \, dv \geq \frac{1}{2} \left[ \psi \left( \frac{1}{2} \right) - \ln 2\pi \right].$$

Based on Theorem 1, we have that when

$$\int_{\mathbb{R}^2} |W_F(\tau, v)|^2 (\tau^2 + v^2)^{-\frac{\alpha}{2}} d\tau dv = D_\alpha \int_{\mathbb{R}^2} |A_F(t, u)|^2 (t^2 + u^2)^{\frac{\alpha}{2}} dt du,$$

the inequality becomes an equality.  $\square$

Third, we further derive the logarithmic uncertainty relation for the different WVD definition (WDL) associated with the LCT proposed in [4]. This states that, if the signal is an odd or even function and its energy is 1, then the logarithmic integral of the WDL is not less than a value with parameter  $b$ . The result is dependent on the value of  $a$ .

**Theorem 4** Suppose that  $f(t) \in L^2(\mathbb{R})$  and  $\|f\|_2 = 1$ .  $W_A^f$  is the WVD associated with the LCT proposed in [4]. If  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , then the following inequality holds:

(1) if  $a \neq 0$ , then

$$\int_{\mathbb{R}^2} |W_{A_1}^f(x, y)|^2 \ln \sqrt{x^2 + y^2} dx dy + \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 \ln \sqrt{u^2 + v^2} du dv \geq D$$

and when

$$C_\alpha \int_{\mathbb{R}^2} |W_{A_1}^f(u, v)|^2 (u^2 + v^2)^{\frac{\alpha}{2}} du dv = \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2)^{-\frac{\alpha}{2}} du dv$$

this inequality becomes an equality.

(2) if  $a = 0$ , then

$$\int_{\mathbb{R}^2} |W_A^f(u, v)|^2 \ln \sqrt{u^2 + v^2} du dv \geq 2\pi^2 b^2 \left( \psi\left(\frac{1}{2}\right) - \ln \pi \right) + \frac{1}{2} \ln \pi |b|$$

and when

$$C_\alpha \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2)^{\frac{\alpha}{2}} du dv = \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2)^{-\frac{\alpha}{2}} du dv$$

this inequality becomes an equality. Where  $D = 4\pi^2 b^2 (\psi(\frac{1}{2}) - \ln \pi) + \ln \pi |b|$ ,

$C_\alpha = 4^a (\pi b)^a D_\alpha$ ,  $0 \leq \alpha < 2$ , and  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ ,  $A_1 = \begin{pmatrix} \frac{a}{4} & b \\ c & 4d \end{pmatrix}$ .

**Proof** From the definition of the AFL, we find that

$$AFL_A^f(2\tau, 2u) = \frac{1}{2\sqrt{i2\pi b}} \int_{\mathbb{R}} f\left(\tau + \frac{t}{2}\right) f^*\left(\frac{t}{2} - \tau\right) e^{i\frac{a}{2b}\frac{t^2}{4} - i\frac{1}{b}ut + i\frac{d}{b}(2u)^2} dt.$$

When  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , we have

$$|AFL_A^f(2\tau, 2u)| = \left| \frac{1}{2\sqrt{i2\pi b}} \int_{\mathbb{R}} f\left(\tau + \frac{t}{2}\right) f^*\left(\tau - \frac{t}{2}\right) e^{i\frac{a}{2b}\frac{t^2}{4} - i\frac{1}{b}ut + i\frac{d}{b}(2u)^2} dt \right|.$$

From the definition of the WDL, we find that if  $a = 0$ ,

$$2|AFL_A^f(2\tau, 2u)| = |W_A^f(\tau, u)|.$$

Theorem 2 then implies that

$$\begin{aligned} & \int_{\mathbb{R}^2} |AFL_A^f(x, y)|^2 \ln \sqrt{x^2 + y^2} \, dx \, dy + \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv \\ & \geq 4\pi^2 b^2 \left( \psi \left( \frac{1}{2} \right) - \ln \pi \right) + \ln 2\pi |b|. \end{aligned}$$

The left-hand side of the above equation can be rewritten as

$$\begin{aligned} & = \int_{\mathbb{R}^2} \left| \frac{1}{2} W_A^f \left( \frac{x}{2}, \frac{y}{2} \right) \right|^2 \ln \sqrt{x^2 + y^2} \, dx \, dy + \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv \\ & = 2 \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv + \ln 2 \\ & \geq 4\pi^2 b^2 \left( \psi \left( \frac{1}{2} \right) - \ln \pi \right) + \ln 2\pi |b|. \end{aligned}$$

If  $a \neq 0$ , we have

$$\begin{aligned} |AFL_A^f(2\tau, 2u)| & = \left| \frac{1}{2\sqrt{i2\pi b}} \int_{\mathbb{R}} f \left( \tau + \frac{t}{2} \right) f^* \left( \tau - \frac{t}{2} \right) e^{i\frac{a}{2b}t^2 - i\frac{1}{b}ut + i\frac{4d}{b}u^2} \, dt \right| \\ & = \frac{1}{2} |W_{A_1}^f(\tau, u)|, \end{aligned}$$

where  $A_1 = \begin{pmatrix} \frac{a}{4} & b \\ c & 4d \end{pmatrix}$ . Similarly, we have

$$\begin{aligned} & \int_{\mathbb{R}^2} |W_{A_1}^f(x, y)|^2 \ln \sqrt{x^2 + y^2} \, dx \, dy + \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv \\ & \geq 4\pi^2 b^2 \left( \psi \left( \frac{1}{2} \right) - \ln \pi \right) + \ln \pi |b|. \end{aligned}$$

The conditions of the derived inequality become equality can be easily obtained based on the Theorem 2.  $\square$

From Eq. 11, we can see that the WDL can be considered as the LCT of the AFL. Therefore, we can obtain a similar result for the AFL without proof.

**Corollary 2** Suppose that  $f(t) \in L^2(\mathbb{R})$  and  $\|f\|_2 = 1$ .  $AFL_A^f$  is the AF associated with the LCT. When  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , we have the following inequality:

(1) if  $a \neq 0$ , then

$$\begin{aligned} & \int_{\mathbb{R}^2} |AFL_A^f(x, y)|^2 \ln \sqrt{x^2 + y^2} \, dx \, dy + \int_{\mathbb{R}^2} |AFL_{A_2}^f(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv \\ & \geq 4\pi^2 b^2 \left( \psi\left(\frac{1}{2}\right) - \ln \pi \right) + \ln 4\pi |b| \end{aligned}$$

and when

$$C_\alpha \int_{\mathbb{R}^2} |AFL_A^f(u, v)|^2 (u^2 + v^2)^{\frac{\alpha}{2}} \, du \, dv = \int_{\mathbb{R}^2} |AFL_{A_2}^f(u, v)|^2 (u^2 + v^2)^{\frac{-\alpha}{2}} \, du \, dv$$

this inequality becomes equality.

(2) If  $a = 0$ , then

$$\int_{\mathbb{R}^2} |AFL_A^f(u, v)|^2 \ln \sqrt{u^2 + v^2} \, du \, dv \geq 2\pi^2 b^2 \left( \psi\left(\frac{1}{2}\right) - \ln \pi \right) + \frac{1}{2} \ln 4\pi |b|$$

and when

$$C_\alpha \int_{\mathbb{R}^2} |AFL_A^f(u, v)|^2 (u^2 + v^2)^{\frac{\alpha}{2}} \, du \, dv = \int_{\mathbb{R}^2} |AFL_A^f(u, v)|^2 (u^2 + v^2)^{\frac{-\alpha}{2}} \, du \, dv$$

this inequality becomes equality. Where  $C_\alpha = 4^a (\pi b)^a$ ,  $0 \leq \alpha < 2$ , and  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ ,

and  $A_2 = \begin{pmatrix} 4a & b \\ c & \frac{d}{4} \end{pmatrix}$ .

### 3.3 Uncertainty Principle for the WVD

Several logarithmic uncertainty relations associated with the LCT were investigated in Sect. 3.2. In this section, we study the other kind of uncertainty principle associated with the WVD based on the newly derived results in Sect. 3.2. We shall find that, if  $f$  is an odd or even function, the integral of the WVD is not less than some constant.

**Theorem 5** Let  $f(t) \in L^2(\mathbb{R})$  and  $\|f\|_2 = 1$ , and let  $W_f(\tau, u)$  be the WVD of signal  $f(t)$ . If  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , then the following inequality holds:

$$\int_{\mathbb{R}^2} |W_f(\tau, u)|^2 (\tau^2 + u^2) \, d\tau \, du \geq e^{\psi\left(\frac{1}{2}\right) - \ln 2\pi} \quad (19)$$

where  $\psi(t) = \frac{d}{dt} \ln [\Gamma(t)]$ , and  $\Gamma(t)$  is the Gamma function. When

$$\int_{\mathbb{R}^2} |W_f(\tau, v)|^2 (\tau^2 + v^2)^{\frac{-\alpha}{2}} \, d\tau \, dv = D_\alpha \int_{\mathbb{R}^2} |A_f(t, u)|^2 (t^2 + u^2)^{\frac{\alpha}{2}} \, dt \, du,$$

the inequality becomes an equality, where  $D_\alpha = \frac{\pi^\alpha \Gamma^2(\frac{2-\alpha}{4})}{\Gamma^2(\frac{2+\alpha}{4})}$ ,  $0 \leq \alpha < 2$ .

*Proof* From Theorem 1, we have

$$\int_{\mathbb{R}^2} |W_f(\tau, v)|^2 \ln \sqrt{\tau^2 + v^2} \, d\tau \, dv \geq C = \frac{1}{2} \left[ \psi\left(\frac{1}{2}\right) - \ln 2\pi \right].$$

Because the logarithm is a concave function, we can use Jensen's inequality to give

$$\ln \int_{\mathbb{R}^2} |W_f(\tau, u)|^2 (\tau^2 + u^2) \, d\tau \, du \geq \int_{\mathbb{R}^2} |W_f(\tau, u)|^2 \ln(\tau^2 + u^2) \, d\tau \, du.$$

Therefore, we obtain the inequality

$$\int_{\mathbb{R}^2} |W_f(\tau, u)|^2 (\tau^2 + u^2) \, d\tau \, du \geq e^{2C}. \quad (20)$$

Based on the results of Theorem 1, we know that when

$$\int_{\mathbb{R}^2} |W_f(\tau, v)|^2 (\tau^2 + v^2)^{-\frac{\alpha}{2}} \, d\tau \, dv = D_\alpha \int_{\mathbb{R}^2} |A_f(t, u)|^2 (t^2 + u^2)^{\frac{\alpha}{2}} \, dt \, du,$$

the inequality becomes an equality, where  $D_\alpha = \frac{\pi^\alpha \Gamma^2(\frac{2-\alpha}{4})}{\Gamma^2(\frac{2+\alpha}{4})}$ ,  $0 \leq \alpha < 2$ .  $\square$

The above inequality represents the uncertainty relation for the WVD. That is, in the time domain and frequency domain,  $W_f$  cannot be too sharply localized. A similar result can be obtained for the WDL proposed in [4], as represented in the following corollary without proof.

**Corollary 3** Let  $f(t) \in L^2(\mathbb{R})$  and  $\|f\|_2 = 1$ .  $W_A^f$  is the WVD associated with the LCT proposed in [4]. If  $f(t) = f(-t)$  or  $f(t) = -f(-t)$ , then we have the following inequality:

(1) if  $a \neq 0$  then

$$\begin{aligned} & \int_{\mathbb{R}^2} |W_{A_1}^f(x, y)|^2 (x^2 + y^2) \, dx \, dy \cdot \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2) \, du \, dv \\ & \geq e^{8\pi^2 b^2 \left( \psi\left(\frac{1}{2}\right) - \ln \pi \right) + 2 \ln \pi |b|} \end{aligned}$$

and when the following condition satisfied, this inequality becomes equality.

$$C_\alpha \int_{\mathbb{R}^2} |W_{A_1}^f(u, v)|^2 (u^2 + v^2)^{\frac{\alpha}{2}} \, du \, dv = \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2)^{-\frac{\alpha}{2}} \, du \, dv$$

(2) If  $a = 0$  then

$$\int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2) \, du \, dv \geq e^{4\pi^2 b^2 \left( \psi\left(\frac{1}{2}\right) - \ln \pi \right) + \ln \pi |b|}$$

and this inequality becomes equality when the following condition is satisfied.

$$C_\alpha \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2)^{\frac{\alpha}{2}} \, du \, dv = \int_{\mathbb{R}^2} |W_A^f(u, v)|^2 (u^2 + v^2)^{\frac{-\alpha}{2}} \, du \, dv$$

$$\text{Where } C_\alpha = 4^a (\pi b)^a D_\alpha, 0 \leq \alpha < 2, \text{ and } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, A_1 = \begin{pmatrix} \frac{a}{4} & b \\ c & 4d \end{pmatrix}.$$

### 3.4 Practical Signal and Applications

In this subsections, we will discuss the potential issues of the derived results.

First, let a signal  $f(t) = (\frac{a}{\pi})^{\frac{1}{4}} e^{-\frac{at^2}{2}}$ , it is easily to see that  $f$  is odd and  $\|f\|_2=1$ . We then obtain the following WVD from the definition:

$$W_f(t, u) = 2e^{-at^2 - \frac{u^2}{a}}. \quad (21)$$

From Theorem 1, we know that

$$\int_{\mathbb{R}^2} |W_f(t, u)|^2 \ln \sqrt{t^2 + u^2} \, dt \, du \geq \frac{1}{2} \left[ \psi \left( \frac{1}{2} \right) - \ln 2\pi \right]. \quad (22)$$

From the results of Eq. 22, we can easily obtain an estimation of the left-hand side of this equation; however, it is difficult to obtain this just by calculating the integral of this equation.

Second, besides the estimation of the lower bound of integral, the other potential applications of the derived results can be found in filtering and state estimation problems in real applications, such as the methods proposed in [9, 15, 17], this directions are also one of our future research topics.

## 4 Conclusions

Based on the classical logarithmic uncertainty relations in the Fourier domain, we have investigated several logarithmic uncertainty relations for the WVD of the signal  $f(t)$  associated with the LCT. We first derived the logarithmic uncertainty relation related to the classical WVD and then found the related logarithmic uncertainty relations for the LCT. Finally, the other kind of uncertainty principles associated with the WVD and WDL were obtained. These results can be considered as a generalization of the classical results to the LCT domain.

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